Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/29/2014
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Project Title:	Investigation of Microgravity Effects on Physiology, Global Gene Expression, and Biofilm Phenotypes of the Caries-Causing Bacterium Streptococcus mutans		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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Although preventable by fluoridation of water and various dental procedures, dental caries remains one of the most prevalent chronic diseases in the world. The oral bacterium Streptococcus mutans is considered the major etiological agent of dental caries, and displays a remarkable ability to persist as part of dental plaque biofilm on the tooth surface. Traditionally, the approach to maintaining astronaut oral health during space flight missions has been routed in rigorous preventative dentistry. However, in the face of future long-term space flight missions, research and discovery of new methods for maintaining oral hygiene and dental health under a microgravity environment will be required. Recent studies have shown that simulated microgravity exposure can cause increased mandibular and alveolar bone loss and decreased saliva flow, but little research has been dedicated to studying the effects of space flight and microgravity on virulence attributes of oral pathogens such as S. mutans. Therefore, the overall hypothesis of this New Space Biology

Task Description:	Investigation proposal is that S. mutans physiology, biofilm formation, and/or expression of virulence phenotypes will be altered in response to simulated microgravity conditions, and will be tested in two specific aims: (1) Gene expression profiles and cell physiology of S. mutans will be compared when grown in rotating wall vessels to simulate microgravity and "normal" (1×g) gravity conditions, using RNAseq and metabolomics approaches, respectively; (2) S. mutans biofilm formation on hydroxyapatite (HA) and related virulence phenotypes will be compared when grown under simulated microgravity and normal gravity conditions. The outcomes of this pilot study will serve as a springboard to development of a full Ground-Based Research proposal, focusing on understanding the regulatory mechanisms behind microgravity-specific changes in gene expression, physiology, or virulence phenotypes, in the hopes of targeting these for development of future anti-caries therapies that could be easily adapted to space flight. Because this New Space Biology proposal will divulge new knowledge with respect to the effects of microgravity on S. mutans biofilm formation, competence development, and global changes in gene expression and metabolism, it directly addresses the SB Science Plan's Cell, Microbial, and Molecular Biology overarching questions "How do alterations in gravity affect cell and microbial replication, including genetic and metabolic regulation?" (CMM-1a), "How do alterations in gravity affect cell-cell interactions and communication?" (CMM-1c), and "How do alterations in gravity affect microbial biofilm formation and/or altered potential for disease?" (CMM-1d). These studies also relate to priority recommendation P1 ("Establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics") of the NRC 2011 Decadal Survey Report, as well as to the Human Research Roadmap, addressing the "Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities" (ExMC 4.11: Limited dental care capabilities).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although preventable by fluoridation of water and various dental procedures, dental caries remains one of the most prevalent diseases in the world. Caries remains the most common chronic disease among children, and remains untreated in up to 30% of adults over 35 years old. The oral bacterium Streptococcus mutans is considered the major etiological agent of dental caries, and displays a remarkable ability to persist as part of dental plaque biofilm on the tooth surface. Although innovative technologies have not been developed from this research, the results of our research suggest that the caries-causing bacterium Streptococcus mutans is more sensitive to oxidative (hydrogen-peroxide) stress when grown under simulated microgravity conditions. This implies that astronaut dental hygiene in space may benefit from the use of hydrogen-peroxide based toothpastes and mouthwashes. Furthermore, the metabolomics and RNA-seq data collected from S. mutans cultures grown in sucrose-containing media (both in simulated microgravity and normal gravity conditions) may shed light on the bacterial genes and metabolic pathways that promote dental caries development in space and on Earth.
Task Progress:	1. During the first 6 months of the project, we obtained two high-aspect ratio vessel (HARV) systems (one to simulate micro-gravity and one for a normal gravity control) from Synthecon, set up this equipment in our lab, and optimized S. mutans inoculation protocols (including ways to minimize the introduction of air bubbles in the HARVs) and rotation speeds for standard growth media and biofilm growth media experiments. We also tested growth of S. mutans in simulated micro-gravity and normal gravity conditions using several different growth medias (artificial saliva media +/- glucose/sucrose, semi-defined biofilm media with various concentrations of glucose/sucrose, and brain heart infusion broth +/- sucrose), and measured the following growth parameters at different time points (up until 24 hours growth): culture pH, culture optical density at 600 nm (OD600), and culture viability by serial dilution plating to calculate colony forming units per ml (CFU/ml). 2. The bulk of our proposed experiments were performed during the last 6 months of the project. The major experiments and their results are summarized below: a) Based on the preliminary experiments described above, we chose to use semi-defined biofilm media (BM) containing 11 mM glucose and 10 mM sucrose to perform metabolomics and RNA-seq analysis on S. mutans simulated microgravity and normal gravity cultures (proposed in Aim 1 of the original research proposal). This decision was due, in part, in observed structural differences in the way the S. mutans cells formed clumps in the HARVs under each growth condition. b) For mass-spectrometry based metabolomics analysis, we collected S. mutans cell pellet and culture supernatant samples from simulated microgravity HARVs and normal gravity HARVs grown for t=4, 8, and 24 hours. Samples were submitted to Metabolon for metabolomics analysis. These results indicated that very few significant alterations were observed under the experimental conditions in this study. Most of the metabolic alterations occurring in this study likely arose from depletion of carbohydrate sources and decreases in energy metabolism and anabolic growth, most likely reflective of growth phase-transitions. There were subtle differences in the timing of these changes between normal and microgravity cells, which may be reflective of differences in growth rate kinetics or growth potential. c) For RNA-seq analysis of transcriptome alterations between S. mutans microgravity and normal gravity growth, we collected cell pellets from simulated microgravity HARVs and normal gravity HARVs grown for t=8 hours. These samples were submitted to Seqwright genomic services, where rRNA depletion, synthesis of cDNA libraries, Illumina sequencing, read curation, alignment to the published S. mutans genome, and differential expression analysis (including statistics) was performed. These results indicated that expression of 153 genes were up-regulated = 2-fold under microgravity growth, and 94 genes were down-regulated = 2-fold under microgravity growth. In-depth analysis of this dataset is still on-going (including assignment of differentially-expressed genes into functional categories). Preliminary inspection of data revealed that a number of genes involved in stress responses were differentially regulated during microgravity growth. A subset of these gene expression changes have also been validated in our lab by real-time PCR. d) S. mutans cells grown under microgravity conditions displayed altered sensitivity to H2O2 treatment compared to those cultured under normal gravity growth conditions. This pattern held true regardless of the type of growth media used in these experiments. e) The effect of simulated microgravity growth on the ability of S. mutans biofilms to adhere to hydroxyapatite (HA) powder as well as glass microcarrier beads was also tested in these studies. The results from both assays indicated that, at the time point and growth media conditions tested, similar levels of biofilm adherence to substrate were observed between the normal and microgravity HARV cultures. f) Preliminary analysis performed in this research has also revealed that we are able to culture and image (by confocal microscopy) both single species S. mutans biofilms as well as dual-species S. mutans/S. gordonii biofilms.

Bibliography Type:	Description: (Last Updated: 02/19/2026)
Abstracts for Journals and Proceedings	Silva D, Lima G, Matzdorf S, Rice KC. "Investigation of the effects of simulated micro-gravity on Streptococcus mutans physiology and biofilm formation." Presented at the Southeastern American Society for Microbiology Summit (Joint meeting of Florida and Southeastern ASM Branches), Ponte Vedra, FL, September 5-7, 2014. Southeastern American Society for Microbiology Summit (Joint meeting of Florida and Southeastern ASM Branches), Ponte Vedra, FL, September 5-7, 2014. Conference Proceedings distributed at the meeting. , Sep-2014
Articles in Peer-reviewed Journals	Orsini SS, Lewis AM, Rice KC. "Investigation of simulated microgravity effects on Streptococcus mutans physiology and global gene expression." npj Microgravity. 2017 Jan 12;3(1):4. Published online: 12 January 2017. http://dx.doi.org/10.1038/s41526-016-0006-4 ; PubMed PMID: 28649626; PubMed Central PMCID: PMC5460135 , Jan-2017
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